RESEARCH ANALYSIS META TABLE – AUTONOMOUS VECTOR DETECTION AND ELIMINATION

Author	Year	Title	Summary	Strengths and Weaknesses
1. Various (Wiley)	2021	*Deep learning in disease vector image identification*	Uses AI to visually identify mosquitoes from images, aiming to automate vector detection.	Strength: Automates the detection process with high accuracy. Weakness: Relies on vision, which fails at night or with fast-moving targets. Zero elimination capability.
2. Chunduri, U.K.	2023	*Robo Claw: An Autonomous Robotic System for Tracking, Capturing, and Eliminating Insects*	Proposes a robot that autonomously tracks and physically crushes/captures insects.	Strength: Attempts both autonomous tracking AND physical elimination. Weakness: Visual/physical tracking is too slow and impractical for fast, erratic mosquitoes.
3. IAEA	2003	*Disease detection and elimination*	Outlines traditional, large-scale methods for tracking and eliminating vector-borne diseases.	Strength: Highlights the critical global need for vector elimination. Weakness: Exposes the massive lack of autonomy in the field; heavily relies on manual human labor and traps.
4. Abdo, A.	2024	*Drone-Based Mosquito Control: Cloud Dispersal Technologies...*	Uses autonomous drones to fly over targeted areas and drop chemical clouds to kill mosquitoes.	Strength: Achieves autonomous elimination over large areas. Weakness: Completely lacks precise detection ("blind spraying"); harms the environment and beneficial insects.
5. Buzzy et al.	2019	*A benchmarking of learning strategies for pest detection... on tomato plants*	Evaluates how well autonomous farm robots can visually detect pests resting on leaves.	Strength: Proves autonomous robots can hunt pests in real-world edge scenarios. Weakness: Only detects static/resting bugs; lacks an elimination mechanism.
6. Lorenz et al.	2022	*Autonomous mosquito habitat detection using satellite imagery and CNNs...*	Uses AI and satellite maps to autonomously detect where mosquitoes are likely breeding.	Strength: Greatly improves predictive detection at a city-wide scale. Weakness: Only detects habitats (water), not the actual bugs. Cannot eliminate them.
7. Lorenz et al.	2022	*Novel approach to autonomous mosquito habitat detection using satellite imagery...*	Expanded research on using space-based mapping to automate disease risk prediction.	Strength: Removes manual human scouting for breeding grounds. Weakness: Fails to address the immediate detection or elimination of adult, flying disease vectors.
8. Fanioudakis et al.	2018	*Mosquito wingbeat analysis and classification using deep learning*	Uses deep learning on acoustic wingbeat recordings to automatically identify mosquito species.	Strength: Solves the detection problem beautifully using audio instead of vision. Weakness: Strictly a passive software tool; completely lacks the elimination phase.
9. Kiskin et al.	2020	*Bioacoustic detection with wavelet-conditioned CNNs*	Optimizes audio detection AI so it can run autonomously on tiny, cheap field computers (like Raspberry Pis).	Strength: Proves autonomous acoustic detection works in real-time on cheap hardware. Weakness: It only listens and logs data. Does not close the loop to eliminate the threat.

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10. Kiriwanna et al.	2021	*A Parameter-Efficient Cascaded Framework for Mosquito Species ID using MFCC Features*	Uses the 13 MFCC audio features to build a highly efficient, lightweight mosquito identification model.	Strength: MFCCs perfectly solve the real-time, low-power detection problem in the dark. Weakness: Stops at detection. Leaves the elimination problem completely unsolved.